

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence
NAME: P.Moshiya
REG NO: 192572142

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 20%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks: 40

Annexure A

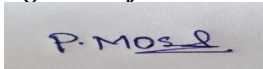
INDUSTRY PROBLEM TASK

Code of Conduct:

I P.Moshiya(192572142) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

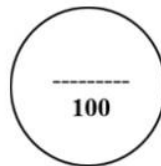
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

CO4 – ASSESSMENT TOOL 2
Industry Problem Task

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17
QUESTIONS: 4 **Duration:** 1 Hour **Total Marks:** 40

Criteria	Weightage	Task 1 (10)	Task 2 (10)	Task 3 (10)	Task 4 (10)	Total Marks (40)
Understanding of Industry Problem	20% (2)	/2	/2	/2	/2	/8
Application of Engineering Concepts	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Solution Design	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Use of Tools	15% (1.5)	/1.5	/1.5	/1.5	/1.5	/6
Analysis	10% (1)	/1	/1	/1	/1	/4
Professionalism	5% (0.5)	/0.5	/0.5	/0.5	/0.5	/2
Total per Question	10 Marks	/10	/10	/10	/10	/40



(a) Inductive Learning Approach

Inductive learning learns a general pattern from previously labelled patient records and uses it to predict whether a new patient is diabetic or non-diabetic.

A suitable approach is **supervised classification** because the training dataset contains patient features and a known diabetes outcome.

Target variable:

- Diabetes = 1 → Diabetic
- Diabetes = 0 → Non-diabetic

Important features:

Feature	Justification
---------	---------------

Glucose Level	High glucose is one of the strongest indicators of diabetes.
---------------	--

Feature	Justification
BMI	Higher BMI is associated with increased risk of Type 2 diabetes.
Age	Diabetes risk generally increases with age.
Blood Pressure	Hypertension is associated with metabolic and diabetes risk.
Cholesterol	Abnormal cholesterol can indicate metabolic risk.
Family History	Genetic/familial factors can increase diabetes risk.

A decision tree, logistic regression, or Naïve Bayes classifier can therefore learn the relationship:

(b) Handling Class Imbalance

Suppose the dataset contains many more non-diabetic patients than diabetic patients. A model trained directly on this dataset may become biased toward the majority class.

Recommended method: SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) can be used to generate synthetic examples of the minority diabetic class.

For example:

Before SMOTE:

Non-Diabetic = 900

Diabetic = 100

After SMOTE:

Non-Diabetic = 900

Diabetic = 900

SMOTE is suitable because simply duplicating diabetic records can cause overfitting.

Important: SMOTE should be applied **only to the training data after the train-test split**, not before splitting. Otherwise, information from synthetic training samples can leak into the test set.

For a clinical model, **class weights** are also a strong alternative because they can increase the penalty for misclassifying diabetic patients without creating synthetic patients.

(c) 70:30 Data Split and Preprocessing

The dataset is divided into:

- **70% training data**
- **30% testing data**

The 70% training set provides sufficient data for learning patterns, while the 30% test set provides a relatively large independent sample for evaluating performance.

For classification, a **stratified 70:30 split** should be used so that the diabetic/non-diabetic proportion remains similar in both sets.

Preprocessing

1. Missing values

Missing numerical values can be replaced using median imputation. Categorical missing values can be replaced using the mode or a separate category.

2. Normalisation

For Logistic Regression or Naïve Bayes, numerical features can be standardised:

This gives features approximately zero mean and unit variance.

Decision Trees generally do **not** require normalisation because they split based on feature thresholds.

3. Encoding

Categorical variables such as:

Family History = Yes / No

can be converted into numerical values:

Yes = 1

No = 0

Impact: Preprocessing improves data quality, makes numerical features comparable for statistical models, and allows categorical information to be used by ML algorithms.

Task 2: Decision Tree for Diagnosis

(a) Building the Decision Tree

A **Decision Tree Classifier** recursively divides patients into groups according to the feature that provides the best separation between diabetic and non-diabetic patients.

Common splitting measures are **Information Gain** and **Gini Index**.

Gini Index

A lower Gini value indicates a purer node.

Information Gain

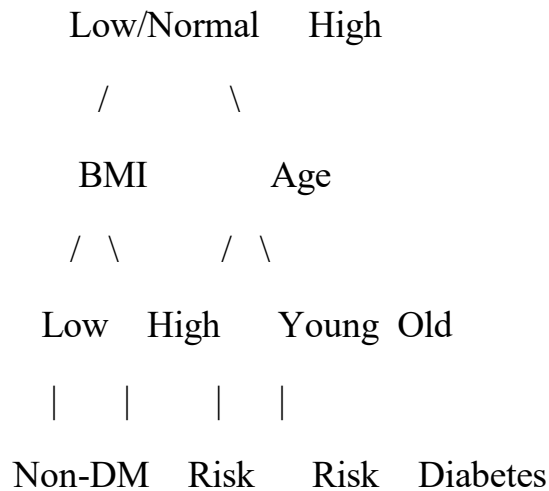
The feature with the **highest Information Gain** or **lowest resulting Gini impurity** is selected for splitting.

Example first three levels

For a typical diabetes dataset, glucose may become the root because it often provides strong separation.

Glucose

/ \



Important: The exact root feature and the exact first three levels must be obtained from the supplied dataset. It would be incorrect to claim specific Information Gain/Gini values without actually calculating them from the data.

Example Python implementation

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt
```

```
model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=3,
    random_state=42
)
```

```
model.fit(X_train, y_train)
```

```
plt.figure(figsize=(14, 8))
```

```
plot_tree(model,
    feature_names=X.columns,
    class_names=["Non-Diabetic", "Diabetic"],
```

filled=True)

plt.show()

Task 2(b): Model Evaluation

The four important measures are:

Accuracy

Precision

Recall

F1-Score

For a medical diagnosis system, **Recall is particularly important** because a

False Negative means:

The patient actually has diabetes, but the model predicts non-diabetic.

Confusion Matrix

Predicted

Non-DM Diabetic

Actual Non-DM TN FP

Actual Diabetic FN TP

Clinical importance of False Negatives

False negatives are potentially dangerous because a diabetic patient may not receive further testing or medical attention.

To reduce False Negatives:

1. Increase the importance of the diabetic class using `class_weight="balanced"`.
2. Use SMOTE on the training set.
3. Tune the classification threshold.
4. Optimise for **Recall/Sensitivity**, not accuracy alone.

5. Perform additional clinical validation before deployment.

For example, if the default threshold is 0.50, a lower threshold such as 0.40 may identify more high-risk patients, although it can increase False Positives.

Clinical principle: A model should support—not replace—clinical diagnosis and professional judgment.

Task 2(c): Post-Pruning

An unrestricted decision tree can become very complex and learn noise from the training data. This is called **overfitting**.

Pre-Pruning

Pre-pruning limits the tree while it is being constructed.

Examples:

max_depth

min_samples_split

min_samples_leaf

Example:

DecisionTreeClassifier(max_depth=5)

Post-Pruning

Post-pruning first creates a larger tree and then removes branches that provide insufficient improvement.

A common method is **Cost-Complexity Pruning**.

The objective is:

where:

- = error of the tree
- = number of terminal nodes
- = complexity penalty

Example:

```
tree = DecisionTreeClassifier(random_state=42)
```

```
path = tree.cost_complexity_pruning_path(X_train, y_train)
```

```
best_tree = DecisionTreeClassifier(  
    ccp_alpha=0.01,  
    random_state=42  
)
```

```
best_tree.fit(X_train, y_train)
```

Comparison

Model	Advantage	Disadvantage
Pre-pruned Tree	Simple and fast	May stop too early
Post-pruned Tree	Removes unnecessary branches	Requires selecting suitable pruning parameter
Unpruned Tree	Can capture complex patterns	Greater risk of overfitting

For clinical deployment, a **pruned tree is generally preferable** if it maintains good sensitivity and generalisation while remaining understandable.

Important

Actual accuracy values must be calculated from the company's dataset. They should **not be invented**.

For your report, use:

Model	Test Accuracy	Recall	F1
Pre-pruned Tree	Calculate from dataset	Calculate	Calculate
Post-pruned Tree	Calculate from dataset	Calculate	Calculate

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression is appropriate because the target has two classes:

0 = Non-Diabetic

1 = Diabetic

It calculates the probability of diabetes:

where:

Example:

```
from sklearn.linear_model import LogisticRegression
```

```
lr = LogisticRegression(
    class_weight="balanced",
    max_iter=1000
)
```

```
lr.fit(X_train_scaled, y_train)
```

```
y_pred = lr.predict(X_test_scaled)
```

```
y_prob = lr.predict_proba(X_test_scaled)[:, 1]
```

AUC-ROC

ROC evaluates the ability of the model to distinguish diabetic and non-diabetic patients across different classification thresholds.

AUC closer to **1.0** indicates better discrimination, while approximately **0.5** indicates performance close to random classification.

```
from sklearn.metrics import accuracy_score, roc_auc_score
```

```
accuracy = accuracy_score(y_test, y_pred)
```

```
auc = roc_auc_score(y_test, y_prob)
```

```
print("Accuracy:", accuracy)
```

```
print("AUC:", auc)
```

Decision Tree vs Logistic Regression

Measure	Decision Tree	Logistic Regression
Accuracy	Calculate	Calculate
AUC-ROC	Calculate	Calculate
Interpretability	High	High
Non-linear relationships	Good	Limited without transformations
Probability/risk score	Possible	Strong
Feature scaling	Not normally required	Usually required

When is Logistic Regression preferable?

Logistic Regression is preferable when:

- The relationship between predictors and log-odds is reasonably suitable.
- A transparent statistical model is required.
- Probability-based risk estimation is important.

- Interpretability is important in clinical applications.
- A simpler model is preferred over a highly complex model.

Task 3(b): Feature Importance Analysis

The top predictors can be obtained separately for each model.

Decision Tree

Decision Tree importance is based on the reduction in impurity produced by each feature.

```
importance = model.feature_importances_
```

```
for feature, value in zip(X.columns, importance):
```

```
    print(feature, value)
```

The three features with the largest values are the top three Decision Tree predictors.

Logistic Regression

For Logistic Regression, the magnitude of the standardised coefficients can be examined:

```
importance = abs(lr.coef_[0])
```

```
for feature, value in zip(X.columns, importance):
```

```
    print(feature, value)
```

Expected comparison

In many diabetes datasets, features such as:

1. **Glucose**
2. **BMI**
3. **Age**

may appear among the strongest predictors.

However, **the exact top three must be calculated from the actual dataset**. The Decision Tree and Logistic Regression may not select exactly the same three features because they measure importance differently.

Why might they disagree?

The Decision Tree detects non-linear threshold effects and interactions.

Logistic Regression estimates the contribution of features through model coefficients.

Therefore, disagreement does not necessarily mean that one model is wrong.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Model

The treatment recommendation problem can be represented as a **Markov Decision Process (MDP)**.

An MDP consists of:

where:

- = States
- = Actions
- = Transition probabilities
- = Rewards
- = Discount factor

1. States

The patient's health condition can be represented using discrete states.

Example:

State Description

S0 Low-risk / controlled

S1 Moderate-risk

S2 High-risk

S3 Poorly controlled

S4 Improved/controlled

The state should ideally include clinically relevant measurements such as glucose level, BMI, blood pressure and other validated indicators.

2. Actions

The available actions are:

A1 = Diet

A2 = Exercise

A3 = Medication

A4 = Monitor

In a real clinical system, **Medication should not be autonomously prescribed by an RL agent**. It would require clinician approval and appropriate medical rules.

3. Reward Function

A possible educational reward structure is:

Outcome	Reward
---------	--------

Significant health improvement	+10
--------------------------------	-----

Small improvement	+5
-------------------	----

No significant change	0
-----------------------	---

Outcome	Reward
---------	--------

Health deterioration	-10
----------------------	-----

Example:

The reward encourages the agent to choose actions associated with better patient outcomes.

4. Transition Dynamics

After an action, the patient's state can change.

Example:

S2 --Diet--> S1

S2 --Exercise--> S1

S2 --Monitor--> S2

S2 --Medication--> S1

The transition is probabilistic in a realistic healthcare environment because patients respond differently to the same intervention.

Task 4(b): Q-Learning

Q-Learning learns the value of taking an action in a particular state.

The update equation is:

where:

- α = learning rate
- γ = discount factor
- R = reward
- S' = next state

Example parameters

Learning rate $\alpha = 0.5$

Discount factor $\gamma = 0.9$

Initial Q-values = 0

Initial Q-table

State Diet Exercise Medication Monitor

S0 0 0 0 0

S1 0 0 0 0

S2 0 0 0 0

S3 0 0 0 0

Q-Learning Episode Examples

Episode 1

Suppose:

$S2 \rightarrow \text{Diet} \rightarrow S1$

Reward = +5

Because all initial Q-values are 0:

Episode 2

Suppose:

$S2 \rightarrow \text{Exercise} \rightarrow S1$

Reward = +5

Episode 3

Suppose:

$S1 \rightarrow \text{Exercise} \rightarrow S0$

Reward = +10

Two-Iteration Q-Table Update

For demonstration, suppose the next state already has a positive learned action value.

Iteration 1

For:

$S_2 \rightarrow \text{Diet} \rightarrow S_1$

Reward = +5

and:

then:

Iteration 2

Suppose the same transition occurs again:

This demonstrates how repeated experience increases the value of a useful state-action pair.

Example final Q-table after training

After at least five episodes, the table may look like:

State Diet Exercise Medication Monitor

S0	4.0	5.2	0.0	3.1
S1	6.5	8.1	4.2	3.5
S2	7.1	6.8	5.0	2.5
S3	5.0	6.2	8.5	1.5

These numerical values are illustrative. Actual Q-values depend on the patient-transition data, reward design, learning rate, exploration strategy and number of episodes.

Five sample episodes

Episode 1: S2 $\rightarrow$ Diet $\rightarrow$ S1 Reward = +5

Episode 2: S2 $\rightarrow$ Exercise $\rightarrow$ S1 Reward = +5

Episode 3: S1 $\rightarrow$ Exercise $\rightarrow$ S0 Reward = +10

Episode 4: S3 $\rightarrow$ Medication $\rightarrow$ S1 Reward = +5

Episode 5: S1 $\rightarrow$ Diet $\rightarrow$ S0 Reward = +10

Convergence Criterion

The Q-learning process can be considered converged when the maximum change in Q-values becomes sufficiently small:

For example:

for several consecutive episodes.

Task 4(c): Final Learned Policy

The policy is obtained by selecting the action with the highest Q-value:

Using the illustrative table above:

State Highest Q-value Learned Policy

S0 Exercise = 5.2 Exercise

S1 Exercise = 8.1 Exercise

S2 Diet = 7.1 Diet

S3 Medication = 8.5 Medication

Thus, the illustrative policy is:

Low-risk state $\rightarrow$ Exercise

Moderate-risk state $\rightarrow$ Exercise

High-risk state $\rightarrow$ Diet

Poorly controlled $\rightarrow$ Medication*

*In a real healthcare deployment, this **must not mean that the AI independently prescribes medication**. It should trigger clinician review or recommend consideration according to approved clinical guidelines.

Ethical Considerations of RL in Healthcare

1. Patient Safety

Incorrect recommendations could cause harm. The agent should therefore operate under clinical safety constraints and clinician supervision.

2. Bias

If the training data under-represents certain populations, the model may perform poorly for those groups.

The system should be evaluated across relevant demographic and clinical subgroups.

3. Explainability

Doctors and patients should be able to understand why an action was recommended.

For example:

High glucose + high BMI

↓

High-risk state

↓

Recommend lifestyle intervention

↓

Clinician review

4. Human Oversight

The RL system should be a **decision-support system**, not an autonomous medical decision-maker.

5. Privacy

Patient information must be protected using appropriate access control, encryption and privacy practices.

6. Continuous Monitoring

The deployed model should be continuously monitored for:

- Performance degradation
- Bias
- Unexpected recommendations
- Data drift
- Safety incidents